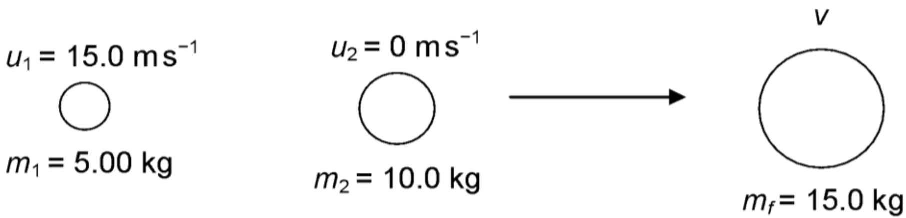
6 **(B)**



Using conservation of momentum,

$$m_1u_1 + m_2u_2 = m_f v$$

$$5.00(15.0) + 10.0(0) = 15.0(v)$$

$$v = 5.00 \text{ m s}^{-1}$$

Loss of kinetic energy = $\frac{1}{2}m_1{u_1}^2 - \frac{1}{2}m_f v^2$

$$= \frac{1}{2}(5.00)(15.0)^2 - \frac{1}{2}(15.0)(5.00)^2$$

$$= 375 \text{ J}$$

7 **(D)**